

Deep Learning Optimizer Visualizer: From SGD to AdamW

A Comprehensive First-Principles Benchmark & Interactive 2D Simulation (100% Pure NumPy)

Project Overview: Interactive visual tool demonstrating deep learning optimization mechanics from scratch. Built with **0% PyTorch/Keras built-in optimizers** and **0% autograd engines**. Features analytical backpropagation, 2D loss surface simulations ($\kappa \in [10, 1000]$), and live MLP benchmarking on real-world datasets.

1. Optimizer Mechanics & Mathematical Update Rules

Optimizer	Mathematical Update Rule	Primary Core Mechanism
SGD	$\theta_{t+1} = \theta_t - \eta g_t$	Direct first-order steepest gradient descent step
Momentum	$v_t = \beta v_{t-1} + (1-\beta) g_t$ $\theta_{t+1} = \theta_t - \eta v_t$	Velocity buffer dampens high-frequency oscillations
NAG	$g_{la} = \nabla L(\theta_t - \beta v_{t-1})$ $v_t = \beta v_{t-1} + (1-\beta) g_{la}, \theta_{t+1} = \theta_t - \eta v_t$	Look-ahead gradient evaluation prevents overshooting
AdaGrad	$G_t = G_{t-1} + g_t^2$ $\theta_{t+1} = \theta_t - [\eta / (\sqrt{G_t} + \epsilon)] \blacksquare g_t$	Parameter-wise adaptive learning rate scaling
RMSProp	$v_t = \beta v_{t-1} + (1-\beta) g_t^2$ $\theta_{t+1} = \theta_t - [\eta / (\sqrt{v_t} + \epsilon)] \blacksquare g_t$	Exponential moving average of squared gradients
Adam	$m_t = m_t / (1-\beta_1^t), v_t = v_t / (1-\beta_2^t)$ $\theta_{t+1} = \theta_t - [\eta / (\sqrt{v_t} + \epsilon)] \blacksquare m_t$	Combines 1st/2nd moments with bias correction
AdamW	$\theta_{t+1} = \theta_t(1 - \eta \lambda) - [\eta / (\sqrt{v_t} + \epsilon)] \blacksquare m_t$	Decoupled weight decay for exact L2 regularization

2. Benchmark Results — Neural Network (Breast Cancer Dataset, 80 Epochs)

Optimizer	Final Train Loss	Final Test Loss	Train Acc.	Test Acc.	Convergence ($\leq 1\%$ Loss)
SGD	0.0657	0.1085	98.90%	95.61%	Epoch 74
Momentum	0.0660	0.1091	98.68%	95.61%	Epoch 74
NAG	0.0653	0.1085	98.90%	96.49%	Epoch 76
AdaGrad	0.0321	0.0790	99.34%	96.49%	Epoch 78
RMSProp	0.0004	0.4696	100.00%	95.61%	Epoch 80
Adam	0.0062	0.2005	100.00%	96.49%	Epoch 80
AdamW	0.0010	0.1406	100.00%	96.49%	Epoch 77

3. Neural Architecture & Quality Assurance

- Neural Architecture:** Fully connected 3-layer MLP ($30 \rightarrow 16 \rightarrow 8 \rightarrow 1$) with He Normal initialization for ReLU hidden layers and Xavier initialization for output Sigmoid layer.
- Loss Function & Gradients:** Binary Cross-Entropy (BCE) loss with analytical backpropagation (zero autograd).
- Unit Test Verification:** 26/26 unit and integration tests passing cleanly (100% pass rate) covering surface Hessians, optimizer updates, reference validation, and Streamlit UI tests.